

Direct Audit of the NS Branching Coefficient in the 111 Channel

Machine working note

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Abstract

We audit the case $n_1 = n_2 = n_3 = 1$ without using the blow-up product to construct the states or to evaluate their matrix element. First we solve for the level-two $\text{Vir} \oplus \text{Vir}$ primary $v_1(P)$ in the auxiliary-fermion times NS-SCA PBW basis. Its five defining highest-weight equations and its ungraded norm are then checked symbolically for generic b, P . We next expand the ordered three-point coefficient into the 25 nonzero auxiliary-fermion contractions, including the full crossing sign printed in the human note, and evaluate every SCA descendant coefficient by Ward identities at $(\infty, 1, 0)$. Only after this direct calculation do we compare with the four- ℓ formula. The norm agrees identically, but the direct 111 three-point numerator does not agree with the printed ℓ -product. The discrepancy is not an overall sign and survives in the normalized square. This note does not change the B^2 used in the double-Virasoro block.

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1 Conventions and scope

We use exactly the parameters of the human note,

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2, \quad h(P) = \frac{Q^2}{8} - \frac{P^2}{2}. \quad (1)$$

The auxiliary fermion is denoted by f_r here (it is denoted by ψ_r in the human note), and

$$f_r^\dagger = -f_{-r}, \quad G_r^\dagger = G_{-r}. \quad (2)$$

The tensor-product pairing is the ungraded one,

$$B_{\text{ug}}(u \otimes x, v \otimes y) = B_F(u, v)B_{\text{SCA}}(x, y). \quad (3)$$

The ordered holomorphic three-point form is nevertheless factorized with the full crossing sign written in the current human note:

$$\widehat{\rho}_a(u_1 \otimes x_1, u_2 \otimes x_2, u_3 \otimes x_3) = (-1)^{|x_1|(|u_2|+|u_3|)+|x_2||u_3|} \rho^F(u_1, u_2, u_3) \rho_a(x_1, x_2, x_3). \quad (4)$$

Thus the pairing convention and the factorization convention are kept distinct throughout the audit.

The notation “111” means $n_1 = n_2 = n_3 = 1$, not the lower case $n_i = \frac{1}{2}$. The total parity is even, so the relevant form is $\widehat{\rho}_0$.

2 The level-two primary $v_1(P)$

2.1 Seven-component ansatz

At total level two and even total parity, the relevant tensor-product PBW basis contains the following seven states:

$$\begin{aligned} &L_{-2}\phi, \quad L_{-1}^2\phi, \quad G_{-\frac{1}{2}}G_{-\frac{3}{2}}\phi, \\ &f_{-\frac{1}{2}}G_{-\frac{3}{2}}\phi, \quad f_{-\frac{1}{2}}G_{-\frac{1}{2}}L_{-1}\phi, \quad f_{-\frac{3}{2}}G_{-\frac{1}{2}}\phi, \quad f_{-\frac{1}{2}}f_{-\frac{3}{2}}\phi. \end{aligned} \quad (5)$$

Define the abbreviations

$$a = \frac{Q}{2} + P, \quad d = (a + b)(a + b^{-1}), \quad R = Q + P. \quad (6)$$

Solving

$$L_m^{(j)}v_1 = 0 \quad (j = 1, 2; \ m = 1, 2), \quad (L_0^{(1)} - h_1^{(1)})v_1 = 0 \quad (7)$$

gives one direction. Fixing its scale by the human-note Fermi-sea normalization gives

$$\boxed{v_1(P) = [4dL_{-2} - 2L_{-1}^2 - 2(a^2 + d)G_{-\frac{1}{2}}G_{-\frac{3}{2}} - 4Rf_{-\frac{1}{2}}G_{-\frac{1}{2}}L_{-1} - 4a^2Rf_{-\frac{1}{2}}G_{-\frac{3}{2}} + 4dRf_{-\frac{3}{2}}G_{-\frac{1}{2}} - 4adRf_{-\frac{1}{2}}f_{-\frac{3}{2}}]\phi(P).} \quad (8)$$

The overall sign can be fixed without a three-point formula. The human note orders

$$w_1 = \chi_{-\frac{3}{2}} \chi_{-\frac{1}{2}} \phi, \quad \chi_r = f_r - i\eta_r. \quad (9)$$

In the canonical auxiliary basis $f_{-\frac{1}{2}} f_{-\frac{3}{2}} \phi$, the pure- f coefficient of w_1 is therefore -1 . Moreover,

$$\ell(Q + 2P, 4) = 16adR. \quad (10)$$

Hence $v_1 = \frac{1}{4}\ell(Q + 2P, 4)w_1$ has pure- f coefficient $-4adR$, exactly as in (8). This removes a possible overall-sign ambiguity before the 111 matrix element is considered.

As an independent source check, equation (B.3a) of [arXiv:1111.2803](#) gives the same seven-component vector after (i) converting to auxiliary-first order, (ii) using $G_{-\frac{1}{2}}^2 = L_{-1}$ and $G_{-\frac{3}{2}} G_{-\frac{1}{2}} = 2L_{-2} - G_{-\frac{1}{2}} G_{-\frac{3}{2}}$, and (iii) multiplying their vector by -2 to obtain the human-note normalization.

2.2 Direct primary check

The code acts on (8) with the two commuting Virasoro generators written in the human note. No ℓ -factor is used in these actions. For symbolic nonzero b, P , the four positive-mode residuals and the $L_0^{(1)}$ residual all simplify identically to zero:

$$\#\{\text{vanishing defining residuals}\} = 5/5. \quad (11)$$

3 Direct ungraded norm

It is useful to split the vector according to its auxiliary state:

$$v_i = Y_i^{(0)} + f_{-\frac{1}{2}} Y_i^{(1)} + f_{-\frac{3}{2}} Y_i^{(3)} + f_{-\frac{1}{2}} f_{-\frac{3}{2}} Y_i^{(13)}, \quad (12)$$

where, for leg i , a, d, R are evaluated at $P = P_i$, and

$$Y^{(0)} = [4dL_{-2} - 2L_{-1}^2 - 2(a^2 + d)G_{-\frac{1}{2}} G_{-\frac{3}{2}}] \phi, \quad (13)$$

$$Y^{(1)} = [-4RG_{-\frac{1}{2}} L_{-1} - 4a^2 RG_{-\frac{3}{2}}] \phi, \quad (14)$$

$$Y^{(3)} = 4dRG_{-\frac{1}{2}} \phi, \quad Y^{(13)} = -4adR\phi. \quad (15)$$

Since $f_r^\dagger = -f_{-r}$, the four auxiliary norms have signs $+, -, -, +$. The ungraded norm is therefore

$$\|v_1\|_{\text{ug}}^2 = \langle Y^{(0)}, Y^{(0)} \rangle - \langle Y^{(1)}, Y^{(1)} \rangle - \langle Y^{(3)}, Y^{(3)} \rangle + \langle Y^{(13)}, Y^{(13)} \rangle. \quad (16)$$

The SCA Gram matrix follows only from

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \quad (17)$$

$$[L_m, G_r] = (\frac{m}{2} - r)G_{m+r}, \quad \{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}(r^2 - \frac{1}{4})\delta_{r+s,0}. \quad (18)$$

Exact symbolic contraction of (16) gives

$$\begin{aligned} \|v_1(P)\|_{\text{ug}}^2 &= \frac{4}{b^6} P(P+b)(Pb+1)(Pb+b^2+1)(2Pb+b^2+1)^2 \\ &\quad \times (2Pb+b^2+3)(2Pb+3b^2+1). \end{aligned} \quad (19)$$

On the other hand,

$$\ell(2P, 4) = 16P(P+b)(P+b^{-1})a, \quad (20)$$

$$\ell(Q+2P, 4) = 16adR. \quad (21)$$

Consequently (19) is exactly

$$\boxed{\|v_1(P)\|_{\text{ug}}^2 = \frac{1}{4} \ell(2P, 4) \ell(Q+2P, 4)}, \quad (22)$$

which is the $n=1$ specialization of the human-note norm formula. This is a generic symbolic identity, not a numerical sample.

4 Direct 111 three-point coefficient

4.1 Fixed-parity SCA three-form

The ordered SCA form is at $(\infty, 1, 0)$ and is normalized by

$$\rho_0(\phi_1, \phi_2, \phi_3) = 1, \quad \rho_1(\phi_1, G_{-\frac{1}{2}}\phi_2, \phi_3) = 1. \quad (23)$$

The eight global values used to terminate the Ward recursion are exactly the two tables of the human note:

$(a_1 a_2 a_3) \mid$	000	110	101	011
$\rho_0 \mid$	1	$h_1 + h_2 - h_3$	$h_1 - h_2 + h_3$	$h_1 - h_2 - h_3$

$(a_1 a_2 a_3) \mid$	100	010	001	111
$\rho_1 \mid$	1	1	-1	$-(h_1 + h_2 + h_3 - \frac{1}{2})$

Here a_i is the power of $G_{-\frac{1}{2}}$ on leg i . In the code this fixed-parity table is obtained from the internal Ward-engine component by the conversion

$$\rho_a^{\text{fixed}}(x_1, x_2, x_3) = (-1)^{a|x_3|} \rho^{\text{engine}}(x_1, x_2, x_3), \quad a = |x_1| + |x_2| + |x_3| \pmod{2}. \quad (24)$$

This conversion is tested against all eight entries above before any level-two coefficient is evaluated.

All remaining SCA descendants in (13)–(15) are reduced recursively by the NS Ward identities. For example, if B, M, K denote the three states, then

$$\rho(B, L_{-1}M, K) = (\Delta_B - \Delta_M - \Delta_K) \rho(B, M, K), \quad (25)$$

and, for $n \geq 2$,

$$\begin{aligned}\rho(B, L_{-n}M, K) &= \sum_{j \geq 0} \binom{n-2+j}{n-2} \rho(L_{n+j}B, M, K) \\ &\quad + (-1)^n \sum_{j \geq 0} \binom{n-2+j}{n-2} \rho(B, M, L_{j-1}K).\end{aligned}\tag{26}$$

For an NS mode $r \in \mathbb{Z} + \frac{1}{2}$, $r > 0$, the middle-leg recursion is

$$\begin{aligned}\rho(B, G_{-r}M, K) &= \sum_{j \geq 0} \binom{r-\frac{3}{2}+j}{j} \rho(G_{r+j}B, M, K) \\ &\quad - (-1)^{|B|+|K|+r+\frac{1}{2}} \sum_{j \geq 0} \binom{r-\frac{3}{2}+j}{j} \rho(B, M, G_{j-\frac{1}{2}}K).\end{aligned}\tag{27}$$

The sums terminate because the external modules are highest-weight modules. Bra modes are moved to the middle and ket legs with the corresponding contour identity; ket modes are reduced by endpoint reflection. Thus every term in the calculation is obtained from the displayed algebra and the eight base values, with no input from the ℓ -product.

4.2 Auxiliary-fermion contractions and crossing signs

Use the labels

$$0 = \mathbf{1}_F, \quad 1 = f_{-\frac{1}{2}}, \quad 3 = f_{-\frac{3}{2}}, \quad 13 = f_{-\frac{1}{2}}f_{-\frac{3}{2}}.\tag{28}$$

Every summand of v_1 is even in total, so its SCA parity equals its auxiliary parity. Therefore the exponent in (4) reduces for the 111 expansion to

$$E = p_1p_2 + p_1p_3 + p_2p_3, \quad p_i = |u_i|.\tag{29}$$

The auxiliary three-form is evaluated directly as the Pfaffian of the sphere fermion two-point function. The bra word is reversed, and the rule $f_r^\dagger = -f_{-r}$ supplies one minus sign per bra fermion. Exactly 25 triples survive:

A	B	C	ρ^F	E	$(-1)^E \rho^F$	A	B	C	ρ^F	E	$(-1)^E \rho^F$
0	0	0	1	0	1	0	1	1	1	1	-1
0	1	3	1	1	-1	0	3	1	-1	1	1
0	3	3	-2	1	2	0	13	13	1	0	1
1	0	1	-1	1	1	1	1	0	-1	1	1
1	1	13	1	1	-1	1	3	13	-2	1	2
1	13	1	1	1	-1	1	13	3	2	1	-2
3	0	3	-1	1	1	3	1	0	-1	1	1
3	1	13	-1	1	1	3	3	0	-1	1	1
3	3	13	1	1	-1	3	13	1	2	1	-2
3	13	3	3	1	-3	13	0	13	1	0	1
13	1	1	-1	1	1	13	1	3	1	1	-1
13	3	1	-1	1	1	13	13	0	1	0	1
13	13	13	-4	0	-4						

Writing $Y_i^{(A)}$ for the SCA state multiplying auxiliary state A on leg i , the direct result is therefore the finite sum

$$T_{111}^{\text{dir}} = \sum_{(A,B,C) \text{ in the table}} (-1)^{E_{ABC}} \rho^F(A, B, C) \rho_0(Y_1^{(A)}, Y_2^{(B)}, Y_3^{(C)}). \quad (30)$$

The implementation also evaluates the seven-by-seven-by-seven expansion without using this table. The unsplit expansion and the 25-block sum agree exactly; this is an internal check on the bookkeeping in (30).

5 Comparison with the human-note ℓ -product

For $n_1 = n_2 = n_3 = 1$, the printed numerator specializes to

$$T_{111}^\ell = -\frac{1}{8}\ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 6\right)\ell\left(\frac{Q}{2} - P_1 + P_2 + P_3, 2\right) \\ \times \ell\left(\frac{Q}{2} + P_1 + P_2 - P_3, 2\right)\ell\left(\frac{Q}{2} - P_1 + P_2 - P_3, -2\right). \quad (31)$$

Since

$$\ell(x, 2) = x, \quad \ell(x, -2) = -\ell(Q - x, 2) = -(Q - x), \quad (32)$$

this becomes

$$T_{111}^\ell = \frac{1}{8}\ell(S, 6)ABC, \quad (33)$$

where

$$S = \frac{Q}{2} + P_1 + P_2 + P_3, \quad A = \frac{Q}{2} - P_1 + P_2 + P_3, \quad (34)$$

$$B = \frac{Q}{2} + P_1 + P_2 - P_3, \quad C = \frac{Q}{2} + P_1 - P_2 + P_3. \quad (35)$$

5.1 Exact one-parameter comparison

To keep the symbolic output readable, take the nonsingular line

$$b = 3, \quad (P_1, P_2, P_3) = \left(-\frac{3}{2}, t, -\frac{3}{2}\right). \quad (36)$$

The direct Ward calculation gives

$$T_{111}^{\text{dir}} = -\frac{(3t+5)}{4251528} (177147t^{11} + 885735t^{10} - 27359370t^9 - 347601780t^8 \\ - 1767367188t^7 - 4991980590t^6 - 7102586493t^5 \\ + 1135874745t^4 + 12286758429t^3 + 1003438035t^2 \\ + 6876067671t + 31288240715). \quad (37)$$

The human-note product instead gives the completely split polynomial

$$T_{111}^\ell = -\frac{1}{157464} t(t+2)(t+8)(3t-4)(3t-2)(3t+4) \\ \times (3t+5)^2(3t+8)(3t+14)(3t+16)(3t+32). \quad (38)$$

Equations (37) and (38) are different polynomials. In particular, their difference is not removed by an overall sign or phase.

At the exact rational point $t = \frac{1}{3}$, all three norms are nonzero and the comparison is

$$T_{111}^{\text{dir}} = -\frac{315962090}{6561}, \quad T_{111}^{\ell} = -\frac{818125}{486}, \quad (39)$$

$$T_{111}^{\text{dir}} - T_{111}^{\ell} = -\frac{609834805}{13122}, \quad \frac{T_{111}^{\text{dir}}}{T_{111}^{\ell}} = \frac{126384836}{4417875}. \quad (40)$$

The three independently verified norms are

$$N_1 = N_3 = \frac{1463}{108}, \quad N_2 = \frac{1971200}{243}. \quad (41)$$

Thus the normalized squares also differ:

$$\frac{(T_{111}^{\text{dir}})^2}{N_1 N_2 N_3} = \frac{998320423171681}{640775109744}, \quad (42)$$

$$\frac{(T_{111}^{\ell})^2}{N_1 N_2 N_3} = \frac{13546875}{7116032}. \quad (43)$$

The second value in (43) is the existing blow-up B^2 used by the double-Virasoro block. It has not been modified by this audit.

As a second generic exact sample, with

$$b = \frac{5}{4}, \quad (P_1, P_2, P_3) = \left(\frac{1}{6}, \frac{2}{7}, -\frac{3}{8}\right), \quad (44)$$

the ratio is

$$\frac{T_{111}^{\text{dir}}}{T_{111}^{\ell}} = \frac{40988240919614376360207863310856861}{3383638222509524104324824472946800} \neq 1. \quad (45)$$

All arithmetic in both samples is exact rational arithmetic.

6 Conclusion and interpretation

The extent of the check is:

1. The explicit seven-component $v_1(P)$ satisfies all five defining primary/eigenvalue equations symbolically for generic b, P .
2. Its ungraded norm agrees symbolically with the $n = 1$ specialization of the human-note norm formula.
3. All 25 nonzero auxiliary-fermion triples are included with $f_r^{\dagger} = -f_{-r}$ and the full crossing exponent in (4).
4. Every SCA coefficient is reduced to the fixed-parity base table by exact Ward identities. The split 25-term sum agrees with a separate unsplit expansion.
5. The direct result disagrees with the printed four- ℓ numerator on an exact one-parameter family and at two explicit rational samples. The disagreement survives in B^2 .

Therefore the following set of statements is not simultaneously consistent: the ungraded tensor pairing, $f_r^\dagger = -f_{-r}$, the full signed factorization (4), the fixed-parity SCA base table, the normalization (8), and the identification of the resulting ordered three-point form with the printed ℓ -product. The primary and norm checks isolate the remaining issue to the dictionary between the ordered $\hat{\rho}$ in (4) and the convention in which the blow-up product was derived. A single additional overall sign cannot repair the 111 result.

This conclusion is deliberately narrower than changing the production branching coefficient. The existing B^2 used in the double-Virasoro block is left intact because it is separately constrained by the block-level comparison with NS c -recursion. What must be repaired is the convention transport that claims that this B^2 is obtained by directly squaring the specific ordered $\hat{\rho}$ defined by the simultaneous conventions above.

7 Reproduction

The exact calculation is implemented in

`Code/audit_ns_branching_111.py.`

Run from the repository root:

```
python3 Code/audit_ns_branching_111.py --compact-line
```

The script constructs (8) directly, performs the generic primary and norm checks, prints the complete auxiliary table, evaluates the two rational samples, and factors the one-parameter polynomials (37) and (38). It imports the existing Ward and Gram engines but does not write to, replace, or patch the production double-Virasoro branching coefficient.